

First year Aerospace Engineering

Mathematics Recap - Algebraic Solving

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Variable isolation - 3 points

Give x in terms of y for the following

1) $2y^2 + 4x = 2(3x - 10)$

2) $4x + y = 3x + 8$

3) $y = \frac{2}{5-x}$

Finding roots - 4 points

Find the roots of the following functions

1) $x^2 - 7x + 10 = 0$

2) $8x - x^2 = 15$

3) $5x^2 + 7x - 9 = 0$

4) $\frac{1}{4} + x^2 = \frac{5}{2}x + \frac{1}{3}x^2$

Systems of linear equations - 4 points

Solve the following systems of linear equations

1) $2x + 3y = 8$

$$5x - 3y = 1$$

2) $4x + 2y = 14$

$$3x - 5y = 22$$

Answers

Slide 2: $x = y^2 + 10, x = 8 - y, x = 5 - \frac{2}{y}$

Slide 3: $x = 5, 2, x = 5, 3, x = \frac{-7 \pm \sqrt{229}}{10}, x = \frac{30 \pm 2\sqrt{201}}{16}$

Slide 4: $(x, y) = (1, 2), (x, y) = (1, 5)$